Offroad Semantic Segmentation

XEN-O-THON 2k26

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MEMBERS:

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PROJECT TITLE:- *Terrain Segmentation using Hybrid AI Models*

"SEGMIFY: Real-Time Safety Equipment Detection Using Synthetic Space Station Data"

PROJECT DELIVERABLES:-

- Trained Segmentation Model
- Dataset & Preprocessing Pipeline
- → Cleaned dataset with mask conversion and augmentation pipeline
- → Hybrid model (DINOv2 + DeepLabV3 +) ready for inference
- Evaluation Results
- → Performance metrics (IoU, Dice Score, Pixel Accuracy) with analysis
- Visualization Outputs
- → Segmented images and comparison results (input vs output)

ABSTRACT:-

Semantic segmentation is one of the most fundamental tasks in computer vision, where each pixel of an image is classified into a predefined category. Unlike traditional classification tasks that assign a single label to an entire image, semantic segmentation provides a dense prediction, enabling a deeper understanding of the scene. This makes it extremely useful in applications such as autonomous driving, robotics, agriculture, and environmental monitoring.

In this project, we propose a deep learning-based semantic segmentation framework that leverages the power of a pre-trained DINOv2 and DeepLabV3 Vision Transformer backbone combined with a ConvNeXt-style segmentation head. The DINOv2 and DeepLabV3 model is used as a feature extractor due to its ability to learn rich representations from large-scale data without requiring labels. The extracted features are then passed through a convolutional segmentation head that reconstructs spatial information and predicts class labels for each pixel.

To improve segmentation accuracy, especially in imbalanced classes, a hybrid loss function combining Cross-Entropy Loss and Dice Loss is used. The performance of the model is evaluated using Intersection over Union (IoU), Dice Score, and Pixel Accuracy. Experimental results demonstrate that the proposed approach achieves reliable segmentation performance on off-road datasets.

INTRODUCTION:-

With the rapid advancement of artificial intelligence and machine learning, computer vision has become a critical area of research. One of the most challenging problems in this field is semantic segmentation, which involves assigning a class label to each pixel in an image. This task requires both local and global understanding of the scene.

Traditional convolutional neural networks (CNNs) have been widely used for segmentation tasks. However, they often struggle to capture long-range dependencies due to their limited receptive field. This limitation leads to inaccurate predictions in complex environments.

Recently, transformer-based architectures have gained popularity due to their ability to model global relationships. DINOv2 and DeepLabV3 are such models that uses self-supervised learning to extract powerful features. It eliminates the need for large labeled datasets and still achieves excellent performance.

In this project, we combine the strengths of transformers and convolutional networks. The DINOv2 and DeepLabV3 backbone extracts features, while a ConvNeXt-inspired head performs segmentation. This hybrid approach ensures both global understanding and efficient computation.

OBJECTIVE:-

The primary goal of this project is to design and implement an efficient semantic segmentation model capable of handling real-world off-road scenarios. The specific objectives include:

- To utilize a pre-trained DINOv2 and DeepLabV3 backbone for feature extraction and reduce training complexity.
- To design a lightweight segmentation head inspired by ConvNeXt architecture.
- To implement a hybrid loss function combining Cross-Entropy and Dice Loss.
- To evaluate the model using multiple performance metrics such as IoU, Dice Score, and Pixel Accuracy.
- To analyze the impact of loss functions on segmentation performance.
- To develop a scalable training pipeline that can run efficiently on GPU platforms such as Kaggle.

These objectives ensure that the model is both accurate and computationally efficient.

Dataset Description:-

The dataset used in this project is an Off-road Segmentation Dataset consisting of RGB images and corresponding segmentation masks. Each mask contains pixel-level annotations representing different terrain elements.

The dataset includes various natural elements such as trees, bushes, rocks, and sky. These classes are essential for applications like autonomous navigation in outdoor environments.

The dataset is divided into:

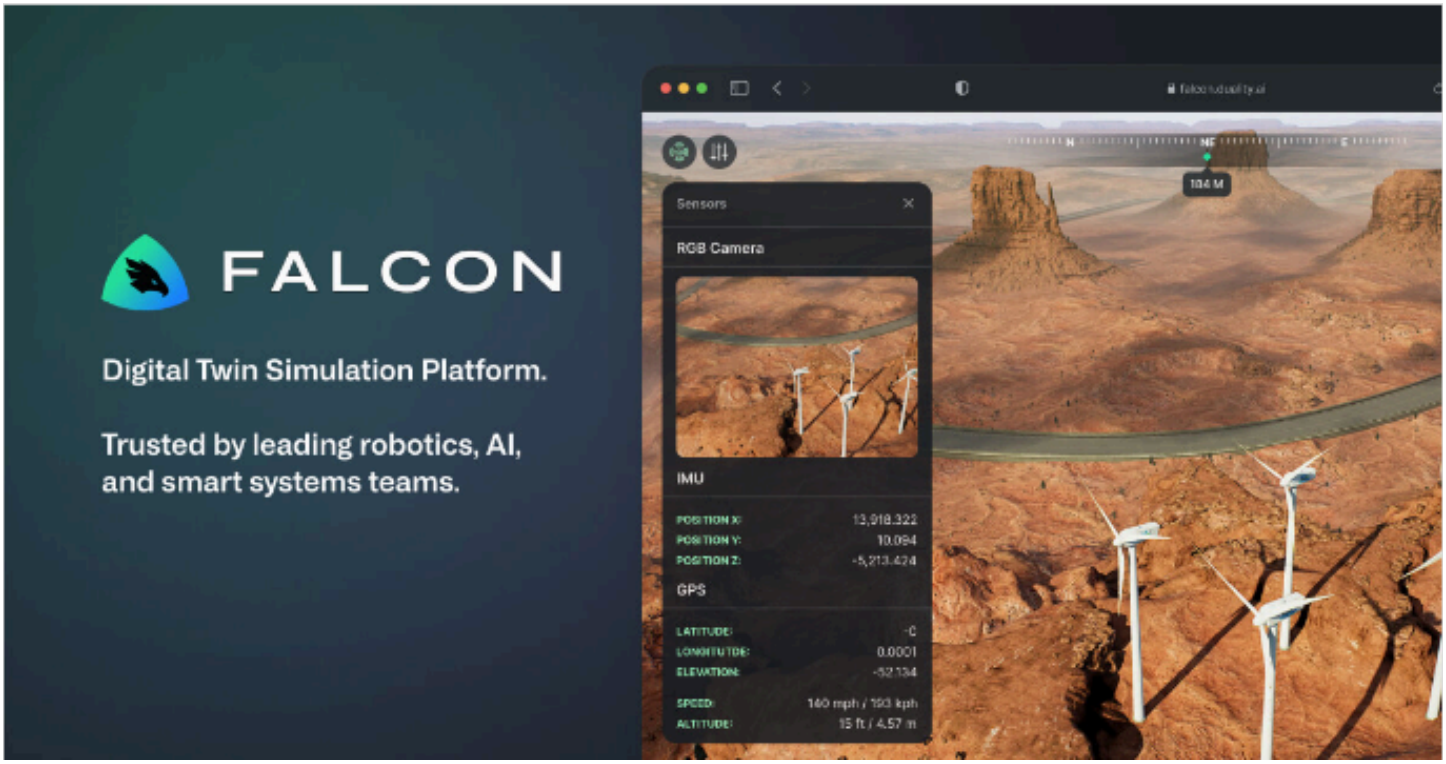
- Training set
- Validation set

Each image has a corresponding mask where pixel values represent class IDs. A mapping function is used to convert raw pixel values into class indices.

Data preprocessing includes:

- Resizing images to a fixed resolution
- Normalizing pixel values
- Converting masks into integer labels

Proper preprocessing ensures stable and efficient training.



Visualization:-

Visualization plays an important role in understanding model performance.

Loss Curves

Loss curves show how training and validation loss decrease over time. A smooth curve indicates stable learning.

IoU Curves

IoU curves demonstrate how segmentation accuracy improves. A rising trend indicates better predictions.

Dice Curves

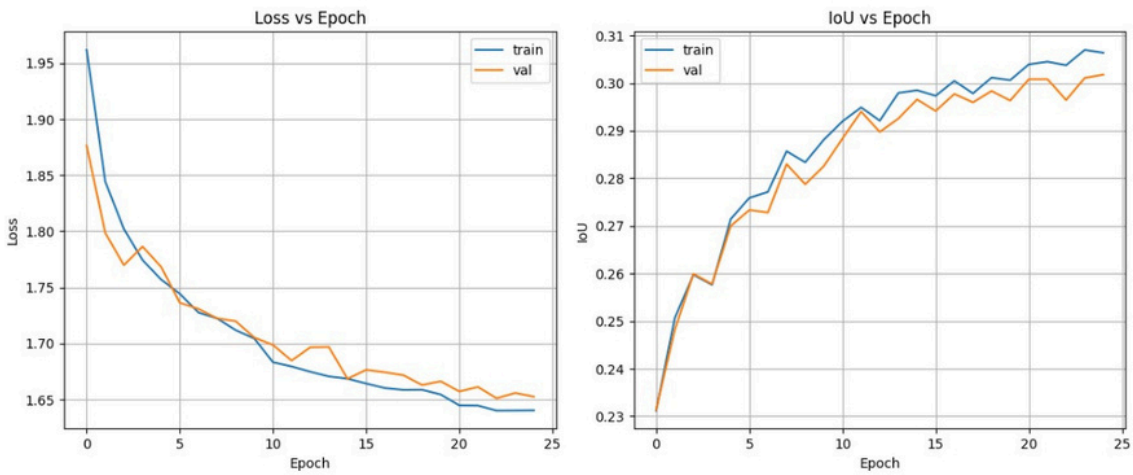
Dice curves highlight improvements in overlap accuracy, especially for small objects.

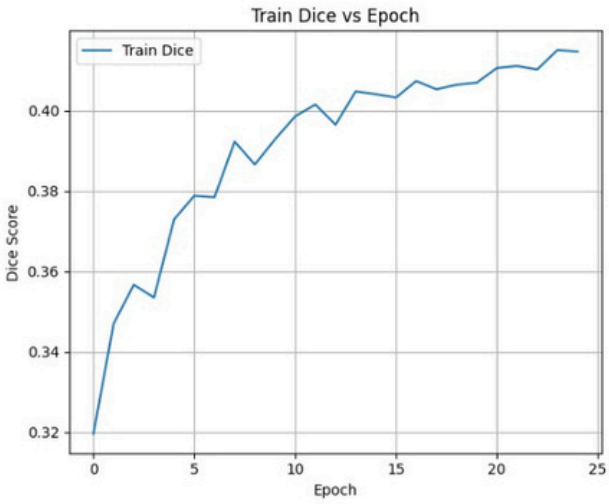
Combined Graph

A combined graph showing all metrics helps compare performance trends.

Importance of Visualization

- Helps detect overfitting
- Provides intuitive understanding
- Aids in debugging




Train Dice vs Epoch
Train Dice
Dice Score
Epoch
0.32
0.34
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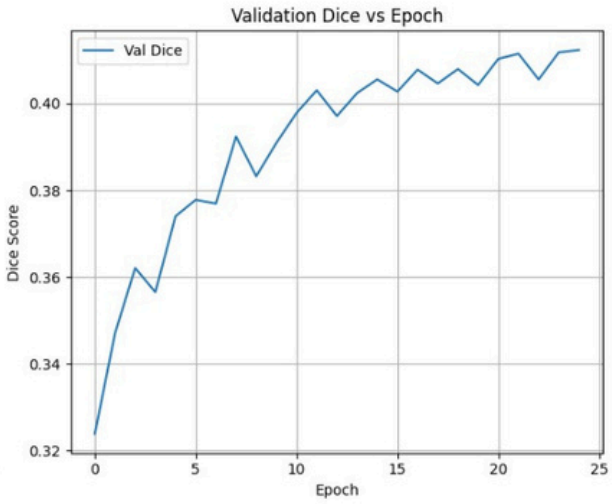
Validation Dice vs Epoch
Val Dice
Dice Score
Epoch
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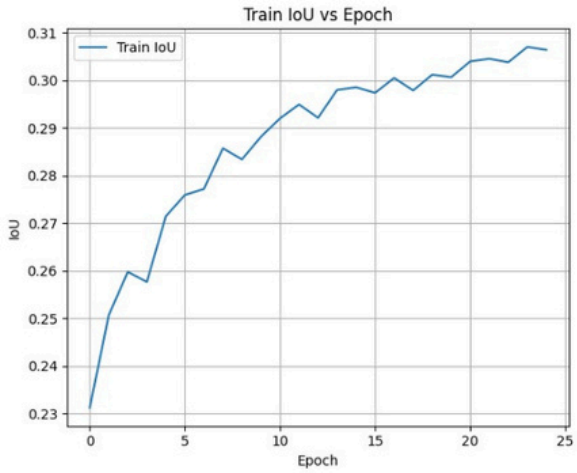
Train IoU vs Epoch
Train IoU
IoU
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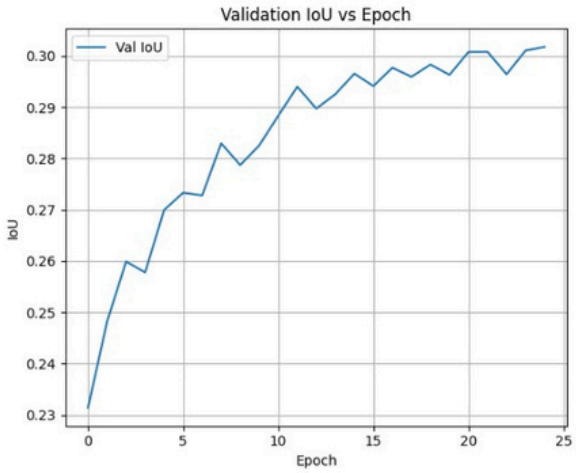
Validation IoU vs Epoch
Val IoU
IoU
Epoch
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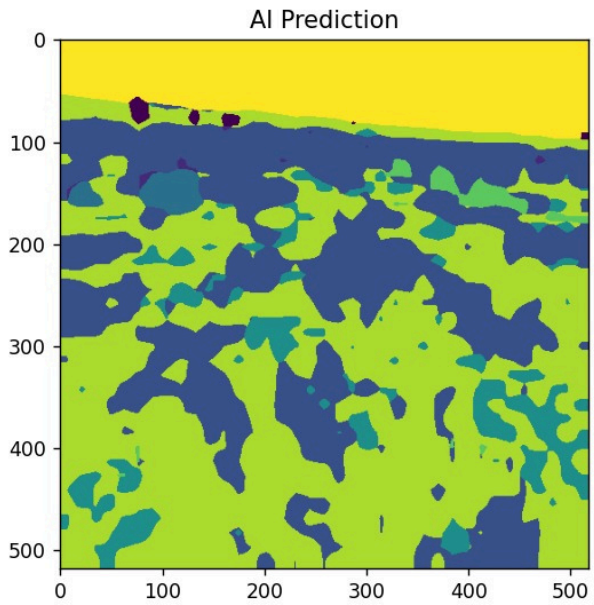
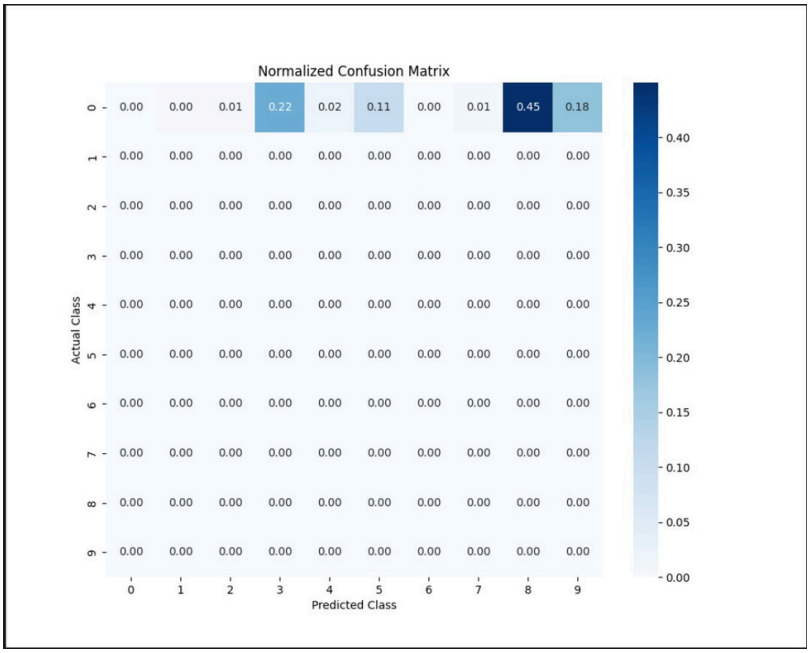
25

NOTED KEY TAKEAWAYS:

1) USE CASE 1:-

MODEL	EPOCH NO.	IOU
DINOV2	10	34
DINOV2 (BACKBONE)+ DEEPLABV3	35	55



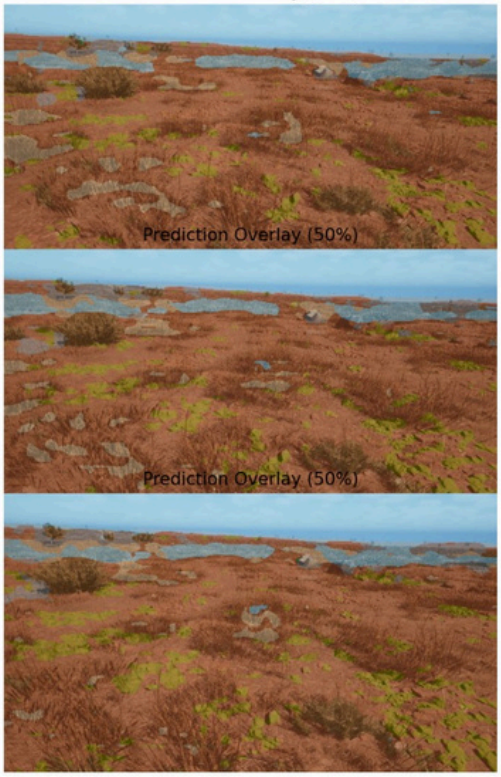


2) USE CASE 2:- BY REDUCING IMAGE SIZE (SLIGHTLY) AND INCREASING VARIANCE

MODEL NO.	EPOCH	IOU
DinoV2101	30	60

3) USE CASE 3 (MAIN RESULT):- REDUCED IN IMAGE SIZE, INCREASE IN VARIANCE , OPTIMIZED LOSS FUNCTION

MODEL NO.	EPOCH	IOU
DinoV2+DeepLabV3	40	60-70
DeepLabV3 Optimised ResNetIO	25	82.5



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✅ All predictions saved locally to: C:\Users\Anshuman Dutta\Downloads\Semantic Dataset\unseen_predictions

Displaying preview of up to 3 samples...
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Advantages:-

The proposed model offers several advantages:

Efficient Feature Extraction

DINOv2 provides powerful features without requiring training from scratch.

Improved Accuracy

Combination of Dice and Cross-Entropy improves segmentation quality.

Lightweight Architecture

ConvNeXt head ensures efficient computation.

Scalability

The model can be extended to larger datasets and more classes.

Limitations:-

Despite its strengths, the model has some limitations:

High Computational Cost:- Transformer models require significant resources.

Dependency on Pre-trained Model:- Performance depends on quality of backbone features.

Limited Real-Time Capability:- Not optimized for real-time applications.

Dataset Dependency:- Results vary based on dataset quality and diversity.

Applications:-

Semantic segmentation has numerous applications:

Autonomous Driving

Helps vehicles understand road and surroundings.

Robotics

Used for navigation and object interaction.

Agriculture

Identifies crops, weeds, and soil conditions.

Environmental Monitoring

Tracks changes in landscapes and ecosystems.

Conclusion:-

This project focuses on semantic segmentation using three different deep learning models:**DINOv2**, **DeepLabV3**, and **SegFormer**. The goal was to analyze and compare their performance in segmenting off-road scenes into multiple meaningful classes.

Model Contributions

- **DINOv2 + ConvNeXt Head**
  - Uses transformer-based feature extraction
  - Captures strong global context
  - Provides high-quality segmentation results with minimal training
- **DeepLabV3**
  - Based on convolutional neural networks
  - Uses atrous (dilated) convolutions for multi-scale feature extraction
  - Performs well on larger objects but struggles with long-range dependencies
- **SegFormer**
  - Hybrid transformer-based model
  - Lightweight and efficient
  - Handles complex scenes and varying object sizes effectively